CultRAG at SemEval-2026 Task 7: Hybrid Sparse-Dense Retrieval with Entity-Centric Knowledge Bases for Cultural MCQ Answering

Aditya Singh* **Rickarya Das***
Indian Institute of Technology Kharagpur
{aditya26189, rickaryadas}@kgpian.iitkgp.ac.in

Abstract

We present a trust-weighted Retrieval-Augmented Generation (RAG; Lewis et al., 2020) system for SemEval-2026 Task 7 (BLEnD) Track 2 (Ousidhoum et al., 2026), targeting English cultural multiple-choice QA across 30 countries. Built atop Llama-3.1-8B-Instruct (Meta AI, 2024), the six-phase pipeline integrates hybrid BM25+FAISS retrieval, country-aware filtering, intent detection, tiered routing, anti-leak prompt engineering, and trust-weighted reranking. The core finding is that RAG *hurts* rather than helps: the LLM-only baseline achieves 78.6% accuracy, outperforming the full system at 78.5% (McNemar's test, $p = 0.962$). Oracle analysis reveals that only 40.7% of questions are answerable from the knowledge base, explaining why retrieval introduces more noise than signal. The sole recovery comes from anti-leak prompt filtering (Phase 4), which mitigates answer-anchoring artifacts. Code: https://github.com/CultRAG/BLEnD-CultRAG.

1 Introduction

The BLEnD benchmark (Myung et al., 2024; Ousidhoum et al., 2026) evaluates LLMs on culturally grounded knowledge across diverse world regions. Cultural knowledge—traditions, social norms, geography, cuisine, local customs—poses a distinctive challenge because LLM training corpora skew Western-centric. Track 2 of SemEval-2026 Task 7 operationalizes this as a four-option MCQ task spanning 47,014 English questions across 30 country-language pairs, requiring cultural understanding unlikely to reside in parametric knowledge alone.

Retrieval-Augmented Generation (RAG; Lewis et al., 2020) is a natural mitigation: a curated cultural KB should fill parametric gaps for underrepresented cultures. We design a six-phase trust-weighted RAG pipeline on this hypothesis.

Our experiments largely reject this hypothesis. The LLM-only baseline (78.6%) matches or exceeds every RAG configuration, and the full system achieves 78.5%—statistically indistinguishable ($p = 0.962$). Oracle analysis shows only 40.7% of questions are answerable from the KB; the LLM's parametric knowledge exceeds the retrieval ceiling—consistent with findings that parametric memory suffices for well-represented knowledge (Mallen et al., 2023). Anti-leak filtering (Phase 4) is the only component producing meaningful recovery, indicating the primary failure mode is answer-anchoring noise, not retrieval quality.

We make four contributions: (1) a six-phase RAG pipeline with trust-weighted reranking and anti-leak filtering; (2) an oracle KB coverage analysis quantifying the theoretical retrieval ceiling; (3) a country-level decomposition revealing that RAG helps low-resource cultures but hurts high-resource ones; and (4) rigorous McNemar significance testing with effect size reporting for all ablation comparisons.

2 System Description

2.1 Knowledge Base Construction

The KB comprises 1,262 text chunks in 7 shards covering all 30 countries, sourced from Wikipedia, travel guides, news outlets, and country portals. Each chunk is assigned a **trust tier**: *High* (Wikipedia, government portals, reference works), *Mid* (reputable travel and news outlets), or *Low* (forums, blogs, unverified sources). Trust tiers drive a multiplicative reranking weight:

$$w_\tau(d) = \begin{cases} 1.0 & \text{if } \tau(d) = \textit{high} \\ 0.6 & \text{if } \tau(d) = \textit{mid} \\ 0.3 & \text{if } \tau(d) = \textit{low} \end{cases} \quad (1)$$

where $\tau(d)$ denotes the trust tier of chunk d (applied in Phase 5, Section 2.5).

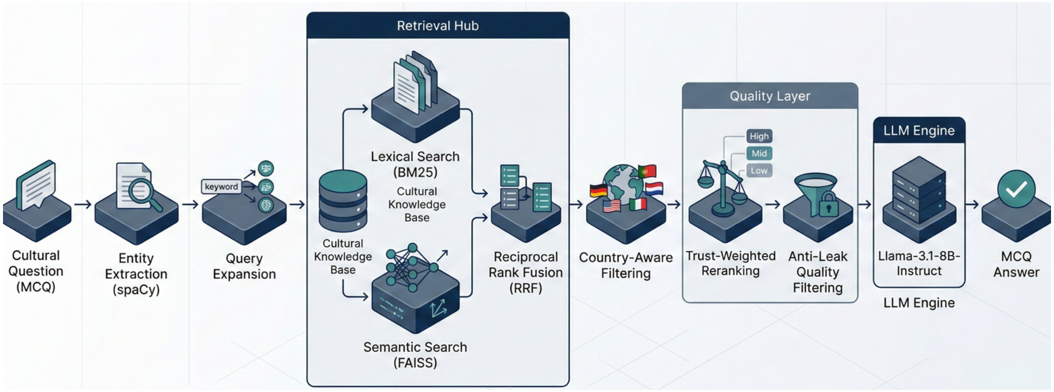

Figure 1: Architecture of the multi-phase cultural RAG pipeline. The system integrates entity extraction, hybrid retrieval (BM25 + FAISS), Reciprocal Rank Fusion, trust-weighted reranking, and phased prompt engineering to generate culturally-grounded answers.

2.2 Entity Extraction

We extract entities using spaCy's (Explosion, 2023) *en_core_web_sm* NER pipeline ($\sim$1.5 entities per question on average: person names, locations, organizations, cultural artifacts). Let $E(q) = \{e_1, \dots, e_m\}$ be the entities from question q. The expanded query is:

$$q' = q \oplus e_1 \oplus \cdots \oplus e_m \tag{2}$$

where $\oplus$ denotes string concatenation, improving recall for chunks mentioning the same entities in variant surface forms.

2.3 Hybrid Retrieval

Let D be the full set of KB chunks, q' the expanded query, and $D_c \subset D$ the subset of chunks tagged with country c. Retrieval operates over D_c.

Lexical Retrieval (BM25). BM25 (Robertson and Zaragoza, 2009) scores each chunk $d \in D_c$:

$$\text{BM25}(q', d) = \sum_{t \in q' \cap d} \text{IDF}(t) \times \frac{f(t,d)(k_1+1)}{f(t,d) + k_1(1 - b + b\frac{|d|}{\text{avgdl}})} \tag{3}$$

where $f(t,d)$ is term frequency, $|d|$ is document length, avgdl is the mean document length over D_c, and:

$$\text{IDF}(t) = \log \frac{|D_c| - n(t) + 0.5}{n(t) + 0.5} \tag{4}$$

with $n(t)$ denoting the number of chunks containing t. We set $k_1 = 1.5, b = 0.75$.

Semantic Retrieval (FAISS). FAISS (Johnson et al., 2021) provides dense semantic retrieval using the *all-MiniLM-L6-v2* sentence encoder (Reimers and Gurevych, 2019) ϕ:

$$\text{Dense}(q', d) = \phi(q')^\top \phi(d) \tag{5}$$

Reciprocal Rank Fusion. The two ranked lists are fused (Cormack et al., 2009):

$$S_{RRF}(d) = \sum_{r \in \{R_{BM25}, R_{Dense}\}} \frac{1}{k + \text{rank}(d, r)} \tag{6}$$

with smoothing constant $k = 60$. The retrieval output is $\mathcal{R}(q') = \text{top-3} \ \text{argmax}_d \, S_{RRF}(d)$. Country filtering restricts candidates to D_c via the question ID prefix (e.g., `ja-JP_0042` $\rightarrow$ Japan), eliminating cross-country contamination.

2.4 Intent Detection

A keyword heuristic classifies questions into 16 intent types (food_drink, sports, government_politics, geography_places, etc.) by matching against question and option text, assigning non-"others" labels to 95.6% of questions (food_drink 46.6%, sports 44.0%, education 4.1%, geography 0.6%, others 4.4%). Despite this coverage, intent labels have zero downstream effect: Phases 2 and 3 produce identical or worse accuracy (Section 2.5), indicating BLEnD does not benefit from intent-conditioned prompting at this granularity.

2.5 Phased Prompt Engineering

The pipeline uses six cumulative prompt engineering phases:

Phase 1 (Country Filter): Restricts retrieved chunks to the target country. Results are identical to unfiltered RAG because entity-expanded queries already steer BM25+FAISS toward country-relevant chunks.

Phase 2 (Intent Detection): Injects intent metadata into the prompt. Despite covering 95.6% of questions (Section 2.4), accuracy is unchanged (77.9%), suggesting the model cannot exploit coarse keyword-derived intent.

Phase 3 (Tiered Routing): Routes questions to intent-specific prompt templates (context-heavy for high-confidence intents, knowledge-light otherwise). Accuracy drops to 77.6%—the worst configuration—indicating poorly calibrated templates add noise rather than structure.

Phase 4 (Anti-Leak Quality Filtering): The sole impactful component. The model is instructed to ignore passage structure, formatting cues, and answer-letter mentions in KB text:

$$P_{final} = \text{Concat}(\text{Prompt}, G(\text{Context})) \qquad (7)$$

The same documents are retrieved; the model reasons independently of surface cues, recovering 1.0pp over Phase 3 to 78.5%.

Phase 5 (Trust-Weighted Reranking): Reorders documents by trust tier before prompt insertion:

$$S_{\text{final}}(d) = S_{RRF}(d) \times w_{\text{tier}}(d), \quad w_{\text{tier}} \in \{1.0, 0.6, 0.3\} \qquad (8)$$

Predictions are identical to Phase 4, suggesting document order has negligible impact at $k = 3$.

Phase 6 (Full System): Combines all phases. The final prediction $\hat{y}$ uses constrained decoding:

$$\hat{y} = \arg \max_{a \in \{A,B,C,D\}} P(a \mid P_{final}, \theta) \qquad (9)$$

where θ denotes the Llama-3.1-8B-Instruct (Meta AI, 2024) parameters. Predictions match Phases 4–5 (36,928 correct), confirming anti-leak filtering is the only component adding value beyond baseline.

3 Experimental Setup

BLEnD Track 2 comprises $N = 47{,}014$ English MCQs (four options each) across 30 country-language pairs with approximately uniform gold labels (A=24%, B=26%, C=26%, D=24%). All experiments use Llama-3.1-8B-Instruct (Meta AI, 2024) on Kaggle GPUs with constrained decoding over $\{A, B, C, D\}$.

Accuracy is the primary metric:

$$\text{Acc} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[\hat{y}_i = y_i^*] \qquad (10)$$

where y_i^* is the gold answer and $\hat{y}_i$ the prediction.

Wilson score confidence intervals:

$$\frac{\hat{p} + \frac{z_{\alpha/2}^2}{2N} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{N} + \frac{z_{\alpha/2}^2}{4N^2}}}{1 + \frac{z_{\alpha/2}^2}{N}} \qquad (11)$$

with $\hat{p} = \text{Acc}$ and $z_{\alpha/2} = 1.96$ for 95% intervals.

Cohen's h measures effect size:

$$h = 2\arcsin\sqrt{p_1} - 2\arcsin\sqrt{p_2} \qquad (12)$$

where p_1, p_2 are the two accuracies being compared.

4 Experimental Results

4.1 Ablation Study

Table 1 summarizes results. The baseline (78.6%) is the best configuration; unfiltered RAG degrades by 0.6pp ($p < 0.0001$). Ph1–Ph2 produce no change (77.9%). Ph3 worsens to 77.6% from poorly calibrated templates. Ph4 is the sole recovery (+1.0pp over Ph3); Ph5–Ph6 add nothing beyond Ph4.

Configuration	Correct	Acc (%)	95% CI	Δ Base
Baseline (LLM only)	36,932	78.6	[78.2, 78.9]	—
RAG Basic (unfiltered)	36,639	77.9	[77.6, 78.3]	−0.6
+ Country Filter (Ph1)	36,639	77.9	[77.6, 78.3]	−0.6
+ Intent Detection (Ph2)	36,639	77.9	[77.6, 78.3]	−0.6
+ Tiered Routing (Ph3)	36,465	77.6	[77.2, 77.9]	−1.0
+ Quality Signals (Ph4)	36,928	78.5	[78.2, 78.9]	−0.0
+ Trust Reranking (Ph5)	36,928	78.5	[78.2, 78.9]	−0.0
Full System (Ph6)†	36,928	78.5	[78.2, 78.9]	−0.0

Table 1: Ablation study results across eight configurations for 47,014 questions. †Official submission.

Figure 2 visualizes the progression: accuracy drops upon adding RAG, stays flat through Phases 1–2, dips at Phase 3, then recovers at Phase 4 to near-baseline.

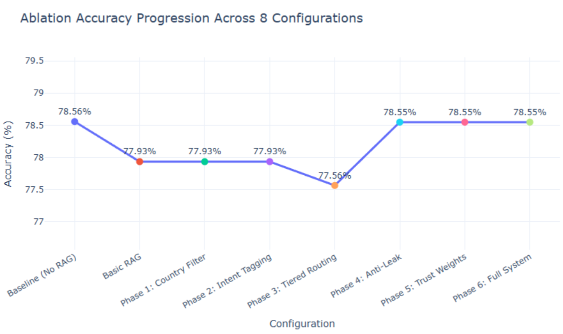

Figure 2: Ablation accuracy progression across eight cumulative pipeline configurations. The baseline (no RAG) achieves the highest accuracy; anti-leak filtering (Phase 4) is the only component that recovers performance.

4.2 Statistical Significance

We use McNemar's test for paired binary outcomes:

$$z = \frac{(|b - c| - 1)}{\sqrt{b + c}} \tag{13}$$

where b and c are discordant pair counts (Table 2). Baseline vs. RAG Basic ($p < 0.0001$) and RAG Basic vs. Full System ($p < 0.0001$) are both significant, but all Cohen's $h \leq 0.015$—detectable at $N = 47{,}014$ yet practically negligible. Baseline vs. Full System is not significant ($p = 0.962$, $h = 0.000$): the full pipeline is indistinguishable from doing nothing.

Comparison	Δ Acc	p-value	Sig	Cohen's h	Effect
Base vs. RAG Basic	-0.6	< 0.0001	Yes	0.015	Small
Base vs. Full System	-0.0	0.962	No	0.000	Small
RAG vs. Full System	$+0.6$	< 0.0001	Yes	0.015	Small

Table 2: McNemar's paired significance tests across 47,014 questions.

5 Analysis

5.1 Why RAG Hurts: KB Coverage Ceiling

We define the **oracle coverage** for a question q_i as:

$$\mathrm{Cov}(q_i) = \mathbf{1}\Big[y_i^* \in \bigcup_{d \in D_c} \mathrm{text}(d)\Big] \tag{14}$$

where y_i^* is the gold answer letter and D_c is the country-specific KB shard. The aggregate coverage ceiling is $\bar{C} = \frac{1}{N}\sum_i \mathrm{Cov}(q_i)$. Only 19,148 questions (40.7%) satisfy $\mathrm{Cov}(q_i) = 1$. The *retrieval ceiling gap* quantifies the futility of retrieval in aggregate:

$$\Delta_{\text{ceil}} = \text{Acc}_{\text{param}} - \bar{C} = 0.786 - 0.407 = +0.379 \tag{15}$$

Coverage varies substantially: Bulgaria (69%), Ecuador (68%), Ethiopia (65%) vs. Great Britain (21%), North Korea (26%), Algeria (26%). Since baseline accuracy (78.6%) exceeds this ceiling, retrieval can only add noise for 59.3% of questions. Great Britain exemplifies the problem: 92.4% baseline vs. 21% coverage, costing 3.0pp under the full system (89.7%). Conversely, Ethiopia (65% coverage, 59.0% baseline) has room for retrieval; the full system gains 4.1pp (63.0%). Figure 3 visualizes this per-country split: retrieval cannot be justified when its coverage ceiling lies below parametric performance.

5.2 RAG Backfire Decomposition

Per-question outcomes form a 2×2 contingency table: a = both correct, b = baseline-only correct, c = system-only correct, d = both wrong; net change $\Delta = c - b$. Baseline vs. RAG Basic: $c = 1{,}810$, $b = 2{,}103$, $\Delta = -293$. Baseline vs. Full System: $c = 1{,}977$, $b = 1{,}981$, $\Delta = -4$ ($p = 0.962$). RAG Basic vs. Full System: Phase 6 recovers 1,479 broken questions while introducing 1,190 new errors (+289 net). Anti-leak filtering mitigates anchoring artifacts without improving retrieval itself.

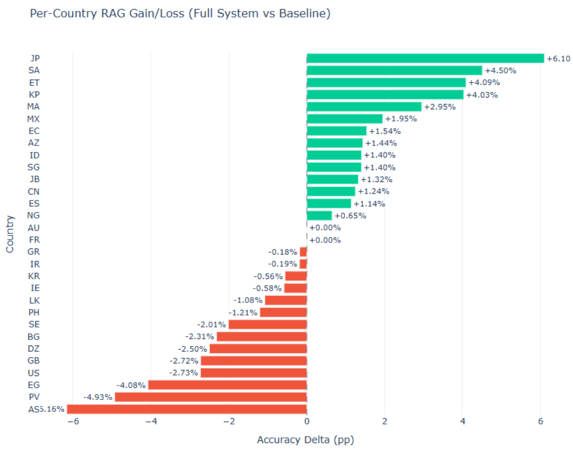

Figure 3: Per-country RAG gain/loss: accuracy delta between the full system (Phase 6) and the LLM-only baseline. Green bars indicate countries where RAG helps; red bars where it hurts.

5.3 Country-Level Analysis

Fourteen countries show improvement: Japan (+6.1pp), Saudi Arabia (+4.5pp), Ethiopia (+4.1pp), North Korea (+4.0pp), Morocco (+2.9pp), Mexico (+1.9pp), Ecuador (+1.5pp), Singapore (+1.4pp), Azerbaijan (+1.4pp), Indonesia (+1.4pp), West Java (+1.3pp), China (+1.2pp), Spain (+1.1pp), and Nigeria (+0.6pp). These are predominantly countries underrepresented in English-language pretraining.

Fourteen countries show degradation: American Samoa (−6.2pp), Basque Country (−4.9pp), Egypt (−4.1pp), Great Britain (−3.0pp), United States (−2.7pp), Algeria (−2.5pp), Bulgaria (−2.3pp), Sweden (−2.0pp), Philippines (−1.2pp), Sri Lanka (−1.1pp), South Korea (−0.6pp), Ireland (−0.6pp), Iran (−0.2pp), and Greece (−0.2pp). These are high-resource countries where parametric knowledge dominates (GB, US), or countries with sparse KB entries where context actively misleads (Basque Country, American Samoa). Figure 4 confirms the pattern: high-baseline countries cluster below the diagonal (RAG hurts); low-baseline countries lie above it (RAG helps). RAG should be applied selectively—conditioned on confidence or coverage—not uniformly.

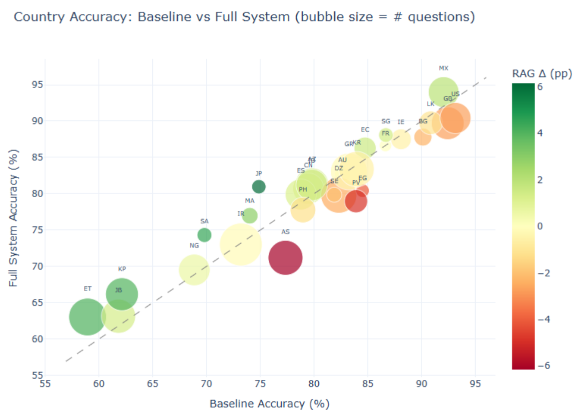

Figure 4: Baseline vs. full system accuracy per country. Bubble size encodes question count; color encodes the RAG delta. Points above the diagonal benefit from RAG.

5.4 Answer Distribution Bias

The baseline over-predicts A (32%) and under-predicts C (22%) and D (21%)—a positional bias common in instruction-tuned LLMs (Wang et al., 2022). Phase 4 partially corrects this (A reduced to ∼30%), though dedicated debiasing (answer shuffling or calibration) would be more effective.

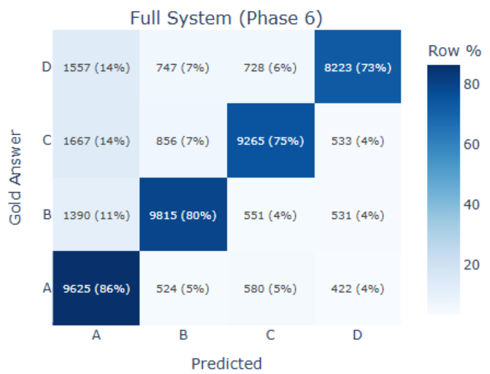

Figure 5: Confusion matrix of full system predictions versus gold labels (47,014 questions). Off-diagonal mass concentrates in the A-predicted row.

5.5 Limitation: Intent-Conditioned Prompting

The keyword classifier covers 95.6% of questions yet yields zero accuracy gain (Section 2.4). The bottleneck is prompt-template design: neither intent metadata injection (Ph2) nor intent-specific routing (Ph3) helps—Ph3 actively hurts. Intent-aware retrieval with a fine-tuned classifier, rather than prompt-only injection, is needed.

6 Official Shared-Task Result

Our submission (CULTRAG, team *aditya26189*) scored **77.55%** accuracy across 29 of 30 BLEnD Track 2 locales, placing 12$^{\text{th}}$ of 18 systems. Locale accuracy ranges from `es-MX` (94.00%) to `am-ET` (63.05%)—a 31-point spread mirroring the coverage gap (Section 5.1). The top five (`es-MX`, `en-US`, `en-GB`, `ta-LK`, `zh-SG`) all exceed 88% and are high-resource or English-native locales. The bottom five (`am-ET`, `su-JB`, `ko-KP`, `ha-NG`, `as-AS`) are low-resource locales where both parametric and retrieved knowledge are sparse.

7 Conclusion

We presented a six-phase trust-weighted RAG pipeline for BLEnD Track 2 atop Llama-3.1-8B-Instruct. The central result is negative: RAG does not help ($p = 0.962$) because parametric accuracy (78.6%) exceeds the KB coverage ceiling ($\bar{C} = 0.407$). Anti-leak filtering is the sole recovery mechanism, gaining 1.0pp by suppressing answer-anchoring artifacts.

Country-level analysis shows RAG benefits low-resource cultures (Japan +6.1pp, Ethiopia +4.1pp) while degrading high-resource ones (American Samoa −6.2pp, Great Britain −3.0pp). Four directions emerge: (1) confidence-conditioned selective retrieval; (2) targeted KB expansion for low-coverage countries; (3) intent-aware retrieval with learned classifiers; and (4) cross-encoder reranking to reduce noise injection.

Ethics Statement

Key ethical considerations: (1) **Systemic Bias**: Wikipedia and WikiVoyage reflect Western-centric perspectives that may misrepresent low-resource cultures. (2) **Cultural Stereotyping**: retrieval may anchor on over-simplified descriptions of traditions. (3) **Potential Misuse**: users might over-rely on the system for cultural advice. Mitigations include **trust-tier weighting** (prioritizing authoritative sources), **anti-leak filtering** (preventing blind context following), and transparent reporting of the RAG Paradox. No personally identifiable information was used.

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